Malaria prevalence and child mortality in Africa — Results summary

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Two complementary analyses relate *Plasmodium falciparum* parasite prevalence (age-standardised $PfPR_{2\text{-}10}$) to all-cause child mortality, and apply the relationship to estimate malaria-attributable child deaths by country and over time. The primary outcome throughout is **post-neonatal under-5 mortality** (1 month–5 years): malaria kills post-neonatal children but essentially no neonates, so neonatal mortality serves as a negative control. All figures/tables are reproduced by the `R/` scripts into `results/`.

Three results:

1. **Deconvolving the IHME model** — parasite prevalence explains the bulk of the between-country variation in malaria's share of child deaths.
2. **A direct survey analysis** — within-country DHS/MIS data reproduce the prevalence–mortality relationship, and a neonatal negative control argues that residual confounding is unlikely to explain it.
3. **Downstream estimates** — malaria-attributable child deaths by country today, and the trajectory since 2000, which falls far faster than WHO/IHME report.

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1. Deconvolution of the IHME model (country level)

For 42 sub-Saharan African countries we computed malaria's **share of child deaths** as GBD/IHME under-5 malaria deaths ÷ IGME all-cause child deaths, and regressed this share on national $PfPR_{2\text{-}10}$ (MAP, 2024), adjusting for log GDP per capita and DTP3 coverage (`R/02_component1_country.R`).

Parasite prevalence is by far the dominant explanator of how large a fraction of a country's child deaths IHME attributes to malaria:

Outcome (malaria's share of…)	PfPR slope (adjusted)	R^2 (PfPR only)	R^2 (+ GDP + DTP3)
post-neonatal deaths	+1.34 pp per PfPR point	0.36	0.54
all under-5 deaths	+0.86 pp per PfPR point	0.43	0.59

Prevalence **alone** accounts for 36–43% of the cross-national variance; adding GDP and DTP3 raises this only to 0.54–0.59 and barely changes the prevalence slope (1.24 to 1.34 and 0.82 to 0.86 pp/point), i.e. the socioeconomic covariates add little once prevalence is known. In effect, IHME's own country-level malaria fractions can be approximated by a simple function of prevalence. One country, **Burundi**, is a clear positive outlier — malaria a much higher share of child deaths than its prevalence predicts (studentised residual > 2).

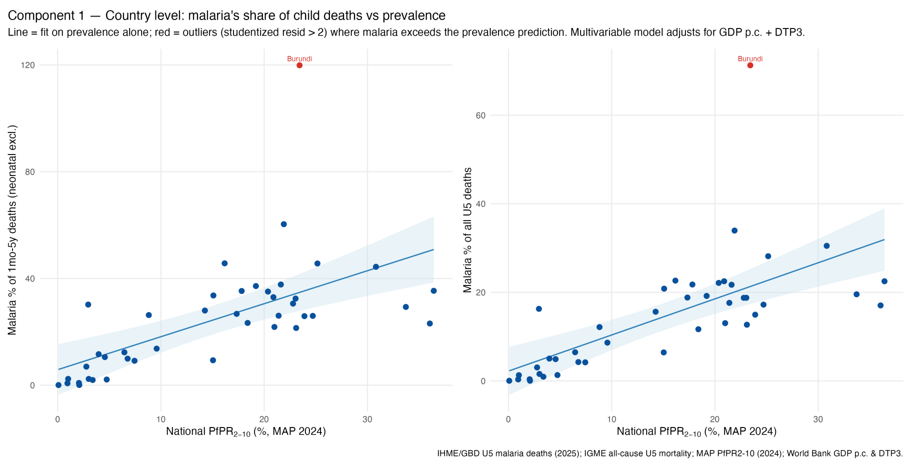

Figure 1: Malaria's share of child deaths vs national prevalence (Method 1)

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2. Direct DHS/MIS survey analysis

The country-level analysis is ecological. We therefore estimated the prevalence–mortality relationship **within** countries, across **467 DHS/MIS survey-regions in 22 countries** with prevalence of 1% or higher (`R/04`, `R/06`, `R/12`). Region child mortality came from the birth histories (DHS.rates synthetic-cohort method); region prevalence was converted to a microscopy footing where needed (microscopy ≈ 0.74 × RDT) and age-standardised to $PfPR_{2\text{-}10}$. The primary model is a negative-binomial GAM for the region death count, **linear**

in PfPR$_{2\text{-}10}$ with mortality on the log scale, a log-exposure offset, a smooth calendar-year term, and country random intercepts and prevalence slopes, adjusted for DTP3, GDP per capita and urbanisation. This linear specification was the best-fitting of four candidate forms (**SText 1**; the penalised prevalence spline collapses to a straight line, edf = 1.0).

Higher prevalence strongly predicts higher post-neonatal mortality: each +10 PfPR$_{2\text{-}10}$ points was associated with a **+8.5% increase in post-neonatal mortality** (95% CI +4.3 to +12.9; $p < 0.001$), an effect that was positive in every country. The implied malaria-attributable fraction (relative to a 1% prevalence counterfactual) rises from ≈ **7% of post-neonatal deaths at 10% prevalence** to ≈ **one third (33%) at 50% prevalence** (and ≈ 48% at 80%).

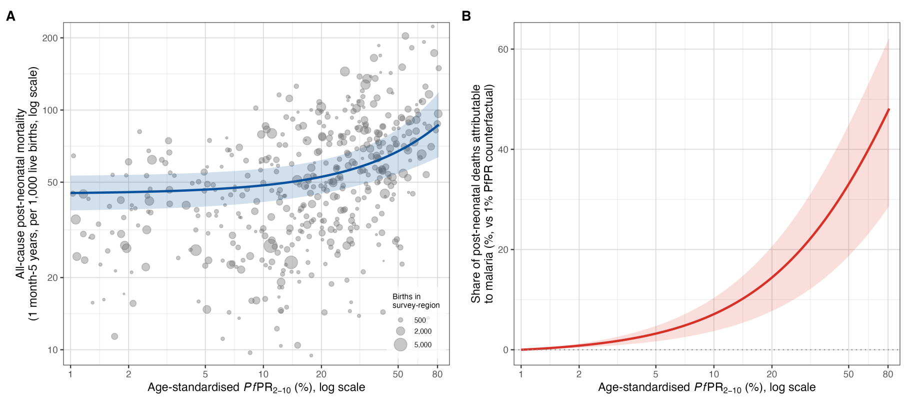

Figure 2: Figure 1: survey-region prevalence vs post-neonatal mortality, and the attributable fraction

Residual confounding is unlikely: the neonatal negative control

If the gradient reflected general socioeconomic confounding rather than malaria, it should also appear for **neonatal** mortality — a period in which malaria causes almost no deaths. It does not. Refitting the identical model by age window:

Age window	% change in mortality per +10 PfPR$_{2\text{-}10}$ pts (95% CI)	p
All under-5	+5.8% (+2.5 to +9.1)	<0.001
Post-neonatal (1mo–5y)	**+8.5% (+4.3 to +12.9)**	<0.001
Neonatal (control)	+0.5% (-2.2 to +3.3)	0.72

The prevalence effect is strong for post-neonatal mortality but **null for neonatal mortality**, exactly as expected if the signal is malaria-specific. This is the central argument against residual confounding: a general “poverty / health-system” proxy would move neonatal mortality too. (At the crude level the neonatal trend is weakly positive — between-country confounding — but adjustment removes it while leaving the post-neonatal signal intact.)

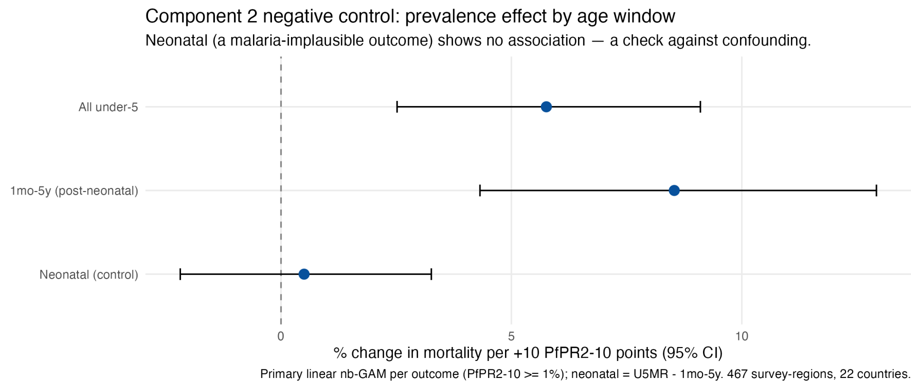

Figure 3: Negative control: prevalence effect by age window

Triangulation with randomised trials

Across cluster-randomised insecticide-treated-net trials that measured both prevalence and child mortality, the observed all-cause mortality reduction per unit prevalence reduction (≈ **8.6% per 10 PfPR$_{\mathbf{2\text{-}10}}$ points removed**; inverse-variance weighted) matches the survey model’s own slope (**8.5%**) — the relationship holds under randomisation, including a zone of the Gambia programme where the intervention failed, prevalence rose, and child mortality rose (`R/10`).

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3. Downstream estimates: deaths by country and over time

Applying the survey attributable fraction (referenced to a 1% counterfactual) to national all-cause child deaths gives malaria-attributable deaths by country (`R/08`) and over 2000–2024 (`R/11`).

National estimates today (25 countries with PfPR$_{\mathbf{2\text{-}10}}$ > 10%)

Malaria-attributable child deaths (thousands; point estimates), Method 2 vs the IHME reference:

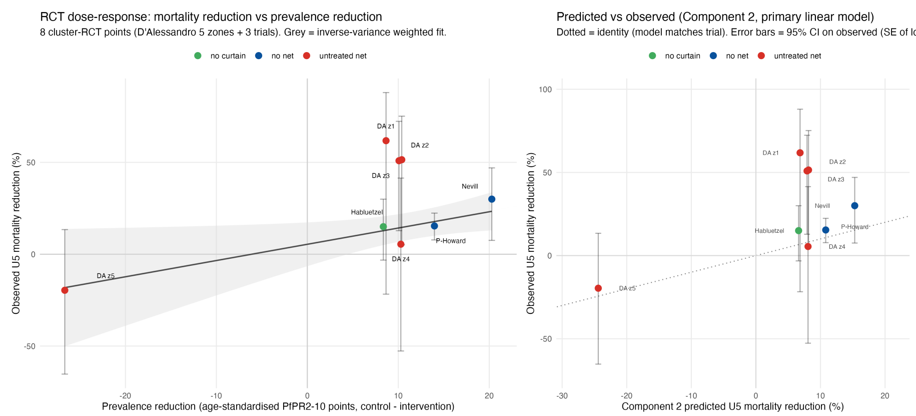

Figure 4: RCT triangulation: predicted vs observed mortality reduction

Country	$PfPR_{2\text{-}10}$	M2 post-neonatal	M2 all-U5	IHME (U5)
Nigeria	24.7%	102.2	108.6	150.5
DR Congo	36.0%	73.7	71.4	68.3
Niger	17.3%	10.9	10.8	23.3
Uganda	21.9%	7.4	9.3	28.5
Mozambique	23.9%	7.5	9.2	11.4
Cameroon	25.2%	7.0	8.0	17.7
Angola	20.4%	6.4	7.1	15.3
Côte d'Ivoire	21.6%	5.8	7.1	14.1
Benin	36.4%	5.8	6.5	8.1
Chad	15.0%	6.4	6.5	5.5
Mali	17.8%	5.6	6.3	15.4
Burkina Faso	20.9%	5.7	5.8	12.4
Guinea	21.4%	4.7	4.9	8.0
South Sudan	23.1%	3.3	3.9	4.3
Central African Rep.	33.7%	3.5	3.7	4.3
Sierra Leone	30.8%	3.5	3.6	7.2
Malawi	19.2%	2.4	3.2	6.3
Ghana	16.2%	1.9	2.6	7.3
Burundi	23.4%	2.2	2.6	15.7
Zambia	14.3%	1.9	2.4	5.3
Togo	21.0%	1.4	1.6	2.0
Liberia	15.1%	1.0	1.1	3.1
Congo-Brazzaville	23.0%	0.7	0.9	1.4
Equatorial Guinea	22.8%	0.4	0.4	0.7
Gabon	18.4%	0.1	0.2	0.3
Total (25)	—	**271** (147–383)	**288** (133–431)	**436**

Applied to national prevalence in these high-burden countries, Method 2's total ($\approx$ 271–288k) sits **below** the summed IHME figure (436k): at today's lower prevalences the direct approach attributes fewer deaths than the cause-of-death model. Nigeria and DR Congo alone are ~60% of the burden. **Burundi** is the largest single divergence (IHME 15.7k vs Method 2 ~2.5k) — the same country flagged as the Method 1 outlier. (Intervals reflect the prevalence-coefficient sampling error only, not uncertainty in prevalence surfaces or all-cause death counts; full per-country CIs in `results/country_malaria_deaths_gt10.csv`.)

The trajectory since 2000

The more consequential result is temporal. Because malaria control lowers prevalence, a method that ties mortality to prevalence registers far more "progress" than cause-of-death models anchored to slowly changing fractions. Summed across sub-Saharan Africa (like-for-like, under-5):

Series	2000	2024	Change
Ours (Method 2, all-U5)	734k	303k	**-59%**
Ours (Method 2, post-neonatal)	766k	283k	-63%
WHO (African region, U5; WMR 2025)	603k	434k	-28%
IHME/GBD (SSA, U5)	564k	428k	-24%

Our estimate starts **higher** than WHO/IHME in 2000 and declines **more than twice as steeply**, crossing the WHO/IHME lines in the mid-2010s. WHO and IHME — two independent systems — agree closely with each other, so the divergence is structural (prevalence-based vs cause-of-death), not an artefact of one dataset. The 2000 level should be read cautiously (the direct approach over-attributes at the very high transmission of that era), but the *steepness of decline* is the key signal: **the reduction in malaria's contribution to child mortality since 2000 has likely been considerably greater than headline figures suggest.**

Methods notes

- **Primary Method-2 model:** linear-in-$\mathrm{PfPR}_{2\text{-}10}$ negative-binomial GAM (log link), analysis sample prevalence 1% or higher (467 regions, 22 countries), attributable fraction referenced to a 1% prevalence counterfactual. Specification choice justified in **SText 1** (linear best-fitting; log-scale spline and power law are sensitivity analyses). Shared fit/AF helpers: `R/00_utils.R` (`fit_c2_primary`, `af_c2`).
- **Prevalence:** MAP $\mathrm{PfPR}_{2\text{-}10}$ (2024 cross-section; 2000–2024 annual rasters for the time series), population-weighted with a gridded population surface. DHS microscopy/RDT prevalence age-standardised to $\mathrm{PfPR}_{2\text{-}10}$ (Smith 2007).

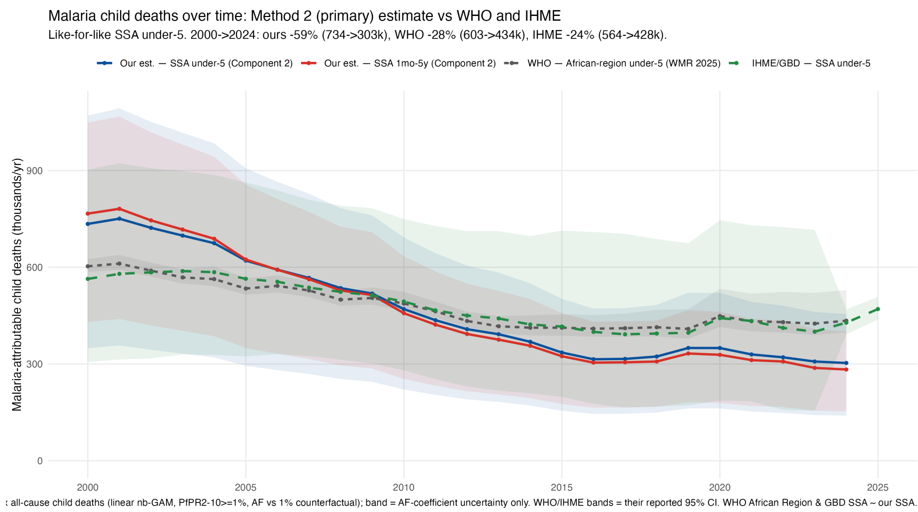

Figure 5: Malaria-attributable child deaths over time vs WHO and IHME

- **Uncertainty bands** here reflect the prevalence-coefficient sampling error only; they are narrower than, and not comparable to, the WHO/IHME intervals.
- **Caveats:** Method 1 is ecological; Method 2 adjusts for the main confounders but is observational, and region-level mortality from a single survey is noisy. MAP $PfPR_{2\text{-}10}$ is 2024 vs the 2025 IHME export (1-year offset). Access-controlled inputs (DHS microdata, IHME/GBD export) are git-ignored under `data/`; code and derived result tables are tracked.